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Studierichting: Informatica
Oefeningenreeks 3D nr. 6

$$f(x) = \frac{2-3x}{x^2-3x+2}$$

$$f'(x) = \frac{-3(x^2-3x+2)-((2-3x)(2x-3)))}{(x^2-3x+2)^2}$$

$$= \frac{-3x^2+9x-6-4x+6+6x^2-9x}{(x^2-3x+2)^2}$$

$$= \frac{x(3x-4)}{((x-2)(x-1))^2}$$

$$f'(x) = 0 \Leftrightarrow \frac{x(3x-4)}{((x-2)(x-1))^2} = 0$$

x	$-\infty$	0		1		$\frac{4}{3}$		2	$+\infty$
$f'(x)$	$+$	0	$-$	$\vert^{(2)}$	$-$	0	$+$	$\vert^{(2)}$	$+$
$f(x)$	$\nearrow$	M	$\searrow$	$\vert$	$\searrow$	m	$\nearrow$	$\vert$	$\nearrow$
		1				9			

$y = ax + b$

$$\left.\begin{array}{l} P(0,1) \in y \\ Q\left(\frac{4}{3}, 9\right) \in y \end{array}\right\} \Leftrightarrow a = \frac{9-1}{\frac{4}{3}} = 6$$

P invullen in $y = 6x + b$

$1 = 6 \cdot 0 + b$

$\Leftrightarrow b = 1$

$\Rightarrow y = 6x + 1$

References